

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12411246
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Renaud; Pierre et al.

Light device of a vehicle

Abstract

A lighting device for a vehicle including at least one light source, at least one sensor configured to acquire information outside the vehicle based on infrared. A cover including a first portion covers the at least one light source and a second portion that is opaque to visible light and to infrared. The cover further includes a third portion that is opaque to visible light and transparent to infrared, substantially centered on said at least one sensor and integrated into the second portion.

Inventors:	Renaud; Pierre (Bobigny, FR), Planche; Gregory (Bobigny, FR), Lozano; Anthony (Bobigny, FR), Joerg; Alexandre (Bobigny, FR)
Applicant:	VALEO VISION (Bobigny, FR)
Family ID:	68281745
Assignee:	VALEO VISION (Bobigny, FR)
Appl. No.:	17/760699
Filed (or PCT Filed):	September 15, 2020
PCT No.:	PCT/EP2020/075782
PCT Pub. No.:	WO2021/052972
PCT Pub. Date:	March 25, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220342079 A1	Oct. 27, 2022

Foreign Application Priority Data

FR	1910197	Sep. 16, 2019
----	---------	---------------

Publication Classification

Int. Cl.: F21S41/20 (20180101); F21S43/20 (20180101); G01S7/481 (20060101); G01S17/931 (20200101)

U.S. Cl.:

CPC G01S17/931 (20200101); F21S41/28 (20180101); F21S43/26 (20180101); G01S7/4813 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2018/0045826	12/2017	Kasaba et al.	N/A	N/A
2018/0229645	12/2017	Hara et al.	N/A	N/A
2018/0231206	12/2017	Hara et al.	N/A	N/A
2019/0126813	12/2018	Hamlin et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1535867	12/2003	CN	N/A
108302434	12/2017	CN	N/A
108413351	12/2017	CN	N/A
108458316	12/2017	CN	N/A
108494909	12/2017	CN	N/A
108534084	12/2017	CN	N/A
0 952 388	12/1998	EP	N/A
3 055 042	12/2017	FR	N/A
3 062 891	12/2017	FR	N/A
2015-40824	12/2014	JP	N/A
2015-43279	12/2014	JP	N/A
WO 2018/124206	12/2017	WO	N/A
WO 2019/058834	12/2018	WO	N/A

OTHER PUBLICATIONS

International Search Report issued on Nov. 19, 2020 in PCT/EP2020/075782 filed on Sep. 15, 2020, 2 pages. cited by applicant

Combined Chinese Office Action and Search Report issued Jan. 30, 2024, in corresponding Chinese Patent Application No. 202080058008.7 (with English Translation of Category of Cited Documents), 8 pages. cited by applicant

Primary Examiner: Baghdasaryan; Hovhannes

Background/Summary

- (1) The present invention relates to a lighting device for a vehicle. It is particularly applicable, but not limited, to motor vehicles.
- (2) In the field of motor vehicles, a lighting device for a vehicle known to those skilled in the art comprises a housing, a light source arranged in the housing, a sensor arranged in the housing and configured to acquire information outside the vehicle based on invisible light, and a cover for closing said housing and comprising a first portion covering the light source and a second portion covering the sensor, in which the second portion has a transmission capacity for visible light that is lower than the first portion. The second portion has a gray or black appearance and also serves an esthetic function.
- (3) One drawback of this prior art is that the second portion which serves an esthetic function therefore covers a large portion of the cover, and the light rays such as infrared rays from outside the vehicle which therefore pass through the second portion, which extends over a large area, degrade elements of the lighting device which are located behind this second portion, such as electronic components or said housing.
- (4) In this context, the present invention aims to provide a lighting device that allows the stated drawback to be remedied.
- (5) To that end, the invention provides a lighting device for a vehicle, comprising at least one light source, at least one sensor configured to acquire information outside the vehicle based on infrared, and a cover comprising a first portion covering said at least one light source and a second portion that is opaque to visible light and to infrared, characterized in that said cover further comprises a third portion that is opaque to visible light and transparent to infrared, substantially centered on said at least one sensor and integrated into the second portion.
- (6) Thus, as will be seen in detail hereinbelow, the third portion, which is integrated into the second portion and centered on the sensor, makes it possible to have a region that allows infrared to pass through but is limited in terms of size. Consequently, the light rays that arrive from outside the vehicle cannot degrade the interior regions of the lighting device since they have only a very small area to pass through the cover.
- (7) According to some non-limiting embodiments, the lighting device may further comprise one or more of the following additional features taken alone or in any technically possible combination, from among the following.
- (8) According to one non-limiting embodiment, said at least one sensor is a lidar.
- (9) According to one non-limiting embodiment, said lighting device comprises a plurality of light sources.
- (10) According to one non-limiting embodiment, said lighting device comprises a plurality of sensors.
- (11) According to one non-limiting embodiment, said cover is made of poly(methyl methacrylate) or of polycarbonate.
- (12) According to one non-limiting embodiment, said third portion comprises a transmission capacity for visible light that is lower than said first portion.
- (13) According to one non-limiting embodiment, said at least one light source is a semiconductor, halogen, xenon, or laser light source.
- (14) According to one non-limiting embodiment, said lighting device is a headlight or a tail light.
- (15) According to one non-limiting embodiment, said second portion and said third portion are covered by said first portion of said cover which is coated over them.

- (16) According to one non-limiting embodiment, said first portion, said second portion and said third portion of said cover are arranged edge to edge.
 - (17) According to one non-limiting embodiment, said first portion, said second portion and said third portion of said cover are injected.
 - (18) According to one non-limiting embodiment, said first portion, said second portion and said third portion are overmolded.
 - (19) According to one non-limiting embodiment, said first portion, said second portion and said third portion are assembled together.
 - (20) According to one non-limiting embodiment, two portions of said first portion, second portion and third portion are overmolded and the other portion of said first portion, second portion and third portion is assembled together with at least one of said two overmolded portions, or vice versa.
 - (21) According to one non-limiting embodiment, said second portion has a transmission capacity for visible light that is lower than said first portion.
 - (22) According to one non-limiting embodiment, said second portion has a transmission capacity for infrared that is lower than said third portion.
 - (23) According to one non-limiting embodiment, said third portion covers said at least one sensor. It is arranged facing said at least one sensor.
-

Description

- (1) The invention and its various applications will be better understood upon reading the following description and with reference to the accompanying figures, in which:
- (2) FIG. 1 schematically illustrates a lighting device for a vehicle, said lighting device comprising a housing, at least one light source, at least one sensor and a cover, according to one non-limiting embodiment of the invention,
- (3) FIG. 2 illustrates a front view of the lighting device of FIG. 1, according to one non-limiting embodiment,
- (4) FIG. 3 illustrates a front view of the housing of the lighting device of FIG. 2, according to one non-limiting embodiment,
- (5) FIG. 4 illustrates a rear view of the cover of the lighting device of FIG. 2, according to one non-limiting embodiment,
- (6) FIG. 5 illustrates a first sectional view of said cover of FIG. 4, according to a first non-limiting embodiment,
- (7) FIG. 6 illustrates a close-up view of a portion of the first sectional view of said cover of FIG. 5, according to one non-limiting embodiment,
- (8) FIG. 7 illustrates a close-up view of a portion of the first sectional view of said cover of FIG. 5, according to one non-limiting embodiment,
- (9) FIG. 8 illustrates a second sectional view of said cover of FIG. 4, according to a second non-limiting embodiment,
- (10) FIG. 9 illustrates a close-up view of a portion of the second sectional view of said cover of FIG. 8, according to one non-limiting embodiment,
- (11) FIG. 10 illustrates a close-up view of a portion of the second sectional view of said cover of FIG. 8, according to one non-limiting embodiment.
- (12) Elements that are identical, in structure or in function, and that appear in several figures use the same reference signs, unless specified otherwise.
- (13) The lighting device 1 for a vehicle 2 is described with reference to FIGS. 1 to 10. In one non-limiting embodiment, the vehicle 2 is a motor vehicle. The term motor vehicle is understood to mean any type of motorized vehicle. This embodiment is taken as a non-limiting example throughout the remainder of the description. Throughout the remainder of the description, the

vehicle is thus also called a motor vehicle.

(14) In some non-limiting examples, the lighting device **1** is a headlight or a tail light.

(15) As illustrated in FIG. **1**, in one non-limiting embodiment, the lighting device **1** comprises: at least one light source **10**, at least one sensor **11**, a cover **12**, a housing **13**.

(16) The lighting device **1** further comprises a device **15** for controlling said at least one light source **10**. In one non-limiting embodiment, the lighting device **1** comprises a plurality of light sources **10**. In one non-limiting embodiment, the lighting device **1** comprises a plurality of sensors **11**.

(17) FIG. **2** illustrates one non-limiting example of a lighting device **1** which is a trunk tail light. It comprises two light sources **10** and two sensors **11**. The sensors **11** are located behind the cover **12**. They are thus illustrated by dashed lines in FIG. **2**.

(18) As illustrated in FIG. **3**, the sensors **11** are positioned in the housing **13**. The housing **13** is closed by the cover **12**. The light source **10** and the sensor **11** are arranged in the housing **13**. The control device **15** is also arranged in the housing **13** as illustrated in the non-limiting example of FIG. **1**. It may be remote in another non-limiting example (not shown).

(19) In one non-limiting embodiment, the light source **10** forms part of an optical module **14** illustrated in FIG. **1**, said optical module **14** further comprising at least one lens and/or reflector **140**. As illustrated in FIG. **1**, the light source **10** is configured to emit light rays R. These light rays R have a wavelength that is in the visible range. The light rays R interact with the lens and/or the reflector **140** of said optical module **14** so as to form all or part of a primary light beam F**1**. Thus, in some non-limiting embodiments, the optical module **14** is configured to form all or part of a signaling beam (such as a daytime running light, position light, fog light, or indicator light), and/or of a lighting beam (such as a high beam or low beam). In one non-limiting embodiment, the light source **10** is a semiconductor light source. In one non-limiting embodiment, the semiconductor light source **10** forms part of a light-emitting diode. What is meant by light-emitting diode is any type of light-emitting diode, i.e., to give a number of non-limiting examples, LEDs (light-emitting diodes), OLEDs (organic LEDs), AMOLEDs (active-matrix organic LEDs) or FOLEDs (flexible OLEDs). In other non-limiting examples, the light source **10** is a halogen, xenon or laser light source.

(20) The sensor **11** is configured to acquire information outside the vehicle **2** based on infrared. In one non-limiting embodiment, the sensor **11** is a lidar. The lidar **11** is configured to emit an infrared light beam F**2** and to receive the echo F**2'** (illustrated in FIG. **1**) in return in order to measure the distance from an object **3** (here a pedestrian in one non-limiting example) located outside the motor vehicle **2**. The echo is the reflection of the infrared light beam F**2** off said object **3**. In one non-limiting embodiment, the lidar **11** measures the travel time between the emitted infrared light beam F**2** and its echo F**2'** which is the reflected light from the infrared light beam F**2**. Thus, it uses the time-of-flight (TOF) technique. It is recalled that infrared has a wavelength from 700 nanometers to 1 millimeter, and the operating wavelength of a lidar is 905 nanometers. From this travel time, in a manner known to those skilled in the art, a distance from the object **3** to the motor vehicle **2** may be determined.

(21) The cover **12** allows the housing **13** to be closed. As illustrated in FIG. **4**, it comprises at least one light region **124** configured to allow the primary light beam F**1** from the light source **10** to pass through. In the non-limiting example taken, it comprises two light regions **124**. As illustrated in FIGS. **5** and **8**, the cover **12** comprises a central portion **12a**, ends **12b** and edges **12c**. It will be noted that the edges **12c** are adjacent on each side to the central portion **12a** of the cover **12** and the ends **12b** are each adjacent to an edge **12c** of the cover **12**. In another non-limiting embodiment (not illustrated), the cover **12** does not comprise ends **12b**, but only a central portion **12a** and edges **12c**. The non-limiting embodiment with the ends **12b** is taken as non-limiting example throughout the remainder of the description.

(22) As illustrated in FIGS. **1**, **5** and **8**, the cover **12** comprises: a first portion **121**, a second portion

122, a third portion **123**.

(23) FIGS. **5** and **8** show sections of FIG. **4** along an axis AA shown in FIG. **4**.

(24) As illustrated in FIG. **1**, the first portion **121** covers the light source **10**. As can be seen in FIG. **1**, the first portion **121** is thus arranged facing the light source **10**. In one non-limiting embodiment, it is placed at a distance from the light source **10**. The first portion **121** is transparent. It allows infrared and visible light to pass through. The primary light beam F1 generated by the optical module **14** passes through the first portion **121** of the cover **12** in order to illuminate a region outside the vehicle **2**. In a first non-limiting embodiment illustrated in FIGS. **1** and **5**, the first portion **121** also covers the sensor **11**. The first portion **121** is arranged at a distance from said sensor **11**. Thus, the infrared light beam F2 passes through the third portion **123** (described later on) as well as the first portion **121**. In a second non-limiting embodiment illustrated in FIG. **8**, the first portion **121** does not cover the sensor **11**. Thus, the infrared light beam F2 passes only through the third portion **123**.

(25) In a first non-limiting embodiment illustrated in FIG. **5**, the first portion **121** extends over the entire length of the cover **12** and covers the entire second portion **122** whether that is in the central portion **12a** of the cover **12** or at the ends **12b** of the cover **12**. In one non-limiting embodiment illustrated, the first portion **121** touches the second portion **122**. In FIGS. **6** and **7**, it is possible to see the first portion **121** covering the second portion **122**. Thus, the first portion **121** covers the second portion **122** and the third portion **123**. In a second non-limiting embodiment illustrated in FIG. **8**, the first portion **121** extends over part of the cover **12** and only partly covers the second portion **122** at the ends **12b** as shown in FIG. **9**. The first portion **121** is adjacent to the second portion **122** in the central portion **12a** of the cover **12** as illustrated in FIGS. **8** and **10**. If the cover **12** does not have ends **12b**, the first portion **121** does not cover the second portion **122**. Thus, the first portion **121** only partly covers the second portion **122** at the ends **12b** of the cover **12** and not the third portion **123**, or does not cover the second portion **122** or the third portion **123**.

(26) In one non-limiting embodiment, the first portion **121** of said cover **12** is made of poly(methyl methacrylate), also known as PMMA, or of polycarbonate, also known as PC. In one non-limiting variant embodiment, when the lighting device **1** is a headlight, the first portion **121** is made of PC. In one non-limiting variant embodiment, when the lighting device **1** is a tail light, the first portion **121** is made of PC or PMMA. The use of PC is explained by its mechanical strength, in particular its impact strength. In addition, when the heat is too high in the rear light, in one non-limiting variant embodiment, the first portion **121** is made of PC.

(27) The second portion **122** acts as an esthetic element allowing the lighting device **1** to be differentiated esthetically from other lighting devices. The second portion **122** is opaque to visible light and to infrared. In one non-limiting embodiment, it may be of a different color. Thus, in some non-limiting examples, it is red, gray or black. In one non-limiting example, when the lighting device **1** is a headlight, it is black in color and when it is a tail light, it is red, gray or black in color. As illustrated in FIG. **1**, the second portion **122** does not cover the sensor **11** or the light source **10**. It is thus not arranged facing the sensor **11**. Nor is it arranged facing the light source **10**. The second portion **122** has a transmission capacity for visible light that is lower than the first portion **121**; i.e. it does not allow visible light to pass through. Additionally, the second portion **122** has a transmission capacity for infrared that is lower than the third portion **123**; i.e. it does not allow infrared to pass through. The primary light beam F1 generated by the optical module **14** thus cannot pass through this second portion **122**. In the same way, the light beam F2 generated by the sensor **11** cannot pass through this second portion **122**. Thus, it filters out visible light and infrared, i.e. their wavelengths. Thus, the second portion **122** which serves an esthetic function and therefore covers a large area of the cover **12** is not dedicated to infrared and therefore is not dedicated to the sensor **11**.

(28) In a first non-limiting embodiment illustrated in FIG. **5**, the second portion **122** is completely covered by the first portion **121** over the entire length of the cover **12** whether that is in the central

portion **12a** of the cover **12** or at the ends **12b** of the cover **12**. In a second non-limiting embodiment illustrated in FIG. 8, the second portion **122** is adjacent to the first portion **121** in the central portion **12a** of the cover **12b** and is covered by the first portion **121** at the ends **12b** of the cover **12**. As illustrated in FIGS. 5 and 8, the second portion **122** extends partly over the central portion **12a** of the cover **12** (except at the location of the third portion **123**) and over the ends **12b** of the cover **12** but not over the edges **12c** of the cover **12** where the light sources **10** are located.

(29) In one non-limiting embodiment, the second portion **122** of said cover **12** is made of poly(methyl methacrylate), also known as PMMA, or of polycarbonate, also known as PC. In one non-limiting variant embodiment, when the lighting device **1** is a headlight, the second portion **122** is made of PC. In one non-limiting variant embodiment, when the lighting device **1** is a tail light, the second portion **122** is made of PC or of PMMA. It will be noted that PC is stronger than PMMA and allows in particular the lighting device **1** (in the case of a headlight) to be protected from gravel which might be projected onto said lighting device **1** when the vehicle **2** is moving, for example. In addition, when the heat is too high in the rear light, in one non-limiting variant embodiment, the second portion **122** is made of PC.

(30) The third portion **123** is transparent to infrared, i.e. it allows infrared light to pass through, unlike the second portion **122**. The third portion **123** is opaque to visible light. It covers the sensor **11**. As can be seen in FIG. 1, the third portion **123** is thus arranged facing the sensor **11**. In one non-limiting embodiment, it is arranged at a distance from said sensor **11**. Thus, the infrared light beam **F2** generated by the sensor **11** passes through this third portion **123** and illuminates a region outside the vehicle **2**. However, like for the second portion **122**, visible light is filtered out by this third portion **123**. When the infrared light beam **F2** reaches an object **3**, the corresponding light **F2'** reflected by the object **3** passes through the third portion **123** and is detected by the sensor **11**. The third portion **123** comprises a transmission capacity for visible light that is lower than the first portion **121**; i.e. the third portion **123** does not allow visible light to pass through. Thus, it filters out visible light, namely its wavelengths. As illustrated in FIG. 1, the third portion **123** is substantially centered on the sensor **11** so as to allow all of the infrared light beam **F2** generated by said sensor **11** to pass therethrough and to be reflected off an object **3** that passes in front of said sensor **11**. The sensor **11** is thus more effective. Furthermore, as illustrated in FIG. 5 and FIG. 8, the third portion **123** is integrated into the second portion **122**. The third portion **123** represents a very limited region of the cover **12** which is thus transparent to infrared. In one non-limiting embodiment, the third portion **123** comprises an area that is substantially equal in size to the area of the sensor **11** to within an integration tolerance **c** for the sensor **11** in the lighting device **1** in order to be sure to completely cover the sensor **11**. In one non-limiting embodiment, the integration tolerance **c** is between a few tenths of a millimeter and one or two millimeters. This makes it possible to take into account the dimensional tolerances for parts, of mechanical assembly sets, etc. In another non-limiting embodiment, in the case where there is a plurality of sensors **11** close to one another, the third portion **123** covers said sensors **11** in order to simplify the design of the cover **12** and improve its ease of mass production. This avoids having a small area of second portion **122** to integrate between the close-together sensors **11**.

(31) Thus, if infrared-heavy light rays come from outside and pass through the third portion **123**, they do not degrade the elements (such as the electronic components or the housing) of the lighting device **1** that are located behind the cover **12**, since the area of the third portion **123** is limited. Thus, the elements are not at risk of being burned. This problem is commonly called "sunburn". Furthermore, since the third portion **123** represents a limited area, the manufacturing cost is reduced. Specifically, the material for obtaining a third portion **123** that is transparent to infrared represents a certain cost. It will be noted that integrating the third portion **123** into the second portion **122** may make it possible to simplify the manufacture of the cover **12**.

(32) In a first non-limiting embodiment illustrated in FIG. 5, the third portion **123** is covered by the first portion **121** when the second portion **122** is entirely covered by the first portion **121**. In one

non-limiting embodiment illustrated, the third portion **123** touches the first portion **121**. In a second non-limiting embodiment illustrated in FIG. **8**, the third portion **123** is not covered by the first portion **121** when the second portion **122** is adjacent to the first portion **121** in the central portion **12a** of the cover **12** and at the ends **12b** of the cover **12**.

(33) In one non-limiting embodiment, the third portion **123** of said cover **12** is made of PMMA or of PC. In one non-limiting alternative embodiment, the third portion **123** is made of PC when the lighting device **1** is a headlight. In another non-limiting alternative embodiment, the third portion **123** is made of PC or of PMMA when the lighting device **1** is a taillight. It will be noted that the material used for the third portion **123** dedicated to the sensor **11** is more expensive than that used for the second portion **122** or the first portion **121**.

(34) The cover is produced according to a manufacturing method described below according to two embodiments.

(35) To produce the cover **12**, a mold (not illustrated) dedicated to the cover **12** is used. The mold is composed of a plurality of cavities in order to produce the cover **12**, each cavity corresponding to a portion of the cover **12**. Thus, the mold comprises a cavity for the first portion **121**, a cavity for the second portion **122** and a cavity for the third portion **123**. The first portion **121**, the second portion **122** and the third portion **123** are injected into their respective cavity.

(36) In a first non-limiting embodiment, the first portion **121**, the second portion **122** and the third portion **123** are overmolded. The cover **12** is thus three-shot injected. Thus, in one non-limiting example, to obtain the first portion **121** which is coated over the second portion **122** and the third portion **123**, the material of the second portion **122** is injected into the cavity of the mold provided for this purpose. Next, the material of the third portion **123** is injected into the cavity of the mold provided for this purpose, or vice versa. Lastly, the material of the first portion **121** is injected into the cavity of the mold provided for this purpose. The first portion coated over the materials of the second portion **122** and of the third portion **123** is thus obtained. Thus, said second portion **122** and said third portion **123** are covered by said first portion **121** which is coated over them. Thus, in another non-limiting example, to obtain the first portion **121**, the second portion **122** and the third portion **123** arranged edge to edge, the material of the first portion **121** is injected into the cavity of the mold provided for this purpose. Next, the material of the second portion **122** is injected into the cavity of the mold provided for this purpose and the material of the third portion **123** is injected into the cavity of the mold provided for this purpose, or vice versa. Injecting the first portion **121** instead of the second portion **122** first avoids having local remelting of the material of the second portion **122** at the border regions which would be too visible.

(37) In a second non-limiting embodiment, the first portion **121**, the second portion **122** and the third portion **123** are assembled together. Thus, in one non-limiting example, the first portion **121**, the second portion **122** and the third portion **123** are arranged edge to edge. In some non-limiting examples, the assembly is performed by bonding, welding, clipping, etc.

(38) In a third non-limiting embodiment, it is possible to combine assembly and overmolding. For example, it is possible to overmold two portions together and assemble a third portion together with one or both of the other portions. Thus, two portions of said first portion **121**, second portion **122** and third portion **123** are overmolded and the other portion of said first portion **121**, second portion **122** and third portion **123** is assembled together with at least one of said two overmolded portions, or vice versa.

(39) Of course, the description of the invention is not limited to the embodiments described above and to the field described above.

(40) Thus, the described invention has the following advantages, in particular: it avoids damage to the elements (electronic components, housing, etc.) located behind the cover **12**, it makes it possible to differentiate the region dedicated to the esthetic function of the cover from the region dedicated to the function of allowing infrared to pass through; thus, unlike the prior art, it does not use the entire area of the second region to let infrared pass through, but rather another area, that of

the third region, to let infrared pass through, it makes it possible to have a smaller region of the cover **12** dedicated to the sensor **12**. It thus reduces the costs for manufacturing of the cover **13**.

Claims

1. A lighting device for a vehicle comprising at least one light source, at least one sensor configured to acquire information outside the vehicle based on infrared, a housing closed by a cover, said cover comprising a first portion covering said at least one light source and a second portion that is opaque to visible light and to infrared, characterized in that said cover further comprises a third portion that is opaque to visible light and transparent to infrared, substantially centered on said at least one sensor and integrated into the second portion.
2. The lighting device as claimed in claim 1, wherein said at least one sensor is a lidar.
3. The lighting device as claimed in claim 1, wherein said lighting device comprises a plurality of light sources.
4. The lighting device as claimed in claim 1, wherein said lighting device comprises a plurality of sensors.
5. The lighting device as claimed in claim 1, wherein said cover is made of poly(methyl methacrylate) (PMMA) or of polycarbonate.
6. The lighting device as claimed in claim 1, wherein said third portion comprises a transmission capacity for visible light that is lower than the first portion.
7. The lighting device as claimed in claim 1, wherein said at least one light source is a semiconductor, halogen, xenon, or laser light source.
8. The lighting device as claimed in claim 1, wherein said lighting device is a headlight or a tail light.
9. The lighting device as claimed in claim 1, wherein said second portion and said third portion are covered by said first portion of said cover which is coated over them.
10. The lighting device as claimed in claim 1, wherein said first portion, said second portion and said third portion of said cover are arranged edge to edge.
11. The lighting device as claimed in claim 1, wherein said first portion, said second portion and said third portion of said cover are injected.
12. The lighting device as claimed in claim 11, wherein said first portion, said second portion and said third portion are overmolded.
13. The lighting device as claimed in claim 11, wherein said first portion, said second portion and said third portion are assembled together.
14. The lighting device as claimed in claim 11, wherein two portions of said first portion, second portion and third portion are overmolded and the other portion of said first portion, second portion and third portion is assembled together with at least one of said two overmolded portions, or vice versa.
15. The lighting device as claimed in claim 2, wherein said lighting device comprises a plurality of light sources.
16. The lighting device as claimed in claim 2, wherein said lighting device comprises a plurality of sensors.
17. The lighting device as claimed in claim 2, wherein said cover is made of poly(methyl methacrylate) (PMMA) or of polycarbonate (PC).
18. The lighting device as claimed in claim 2, wherein said third portion comprises a transmission capacity for visible light that is lower than the first portion.
19. The lighting device as claimed in claim 2, wherein said at least one light source is a semiconductor, halogen, xenon, or laser light source.

20. The lighting device as claimed in claim 2, wherein said lighting device is a headlight or a tail light.
